

estate AI

An AI assistant for commercial property loans — simple guide

1. What is it?

A website that helps a bank check whether a property loan is safe to give.

AI reads the loan details and documents, finds problems, and writes reports — work that usually takes analysts days.

Money ratios (DSCR, LTV, Debt Yield) are calculated by fixed maths, not guessed by AI, so numbers are always correct.

2. Key words (in plain English)

Word	Meaning
NOI	Yearly profit the property makes after its running costs.
DSCR	Profit ÷ yearly loan payments. 1.25x means profit covers payments 1.25 times. Higher = safer.
LTV	Loan as % of property value. 70% means a \$7M loan on a \$10M building. Lower = safer.
Debt Yield	Profit as % of the loan. Higher = safer.
Covenant	A promise the borrower must keep, e.g. "DSCR must stay above 1.20x".
Credit Memo	The final report that tells the bank's committee: approve or decline.

3. How to use it (5 steps)

- **Step 1:** Open *Deal Pipeline* to see all loan requests.
- **Step 2:** Click a deal to open it in the *Underwriting Workspace*.
- **Step 3:** Upload its documents (PDF, scanned PDF or photo) — AI reads them.
- **Step 4:** Ask the Copilot questions and check the risk flags.
- **Step 5:** Open *Credit Memorandum* to get the final approve/decline report.

Every page has a **Regenerate** button — click it to make the AI produce a fresh analysis.

4. Each page — what it does, with an example

① Deal Pipeline & Inbox

Shows every loan request, its stage, priority and how safe it looks.

Example: You see "Harbor View Apartments – \$18.5M – DSCR 1.62x – High priority – 1 blocker". Green DSCR means healthy; you click it to work on it.

② Underwriting Workspace

The main working page for one loan: key numbers, AI-extracted facts, conflicts and conditions.

Example: It shows "NOI \$1.42M (Operating Statement, page 3, 94% confidence)" and warns "Rent roll says 92% occupied, appraisal says 88% — please check".

③ Loan Documents (PDF / Scans) — inside Underwriting

Drop a PDF, scanned PDF or phone photo; AI reads it, pulls out the numbers and compares them to the deal.

Example: You upload a scanned T-12 statement. AI reads it (badge "AI OCR"), finds NOI = \$1.35M instead of \$1.42M, updates the deal, and DSCR is recalculated automatically.

④ Underwriting Copilot — inside Underwriting

A chat box. Ask questions about the deal; AI answers only from the deal's data and shows its sources.

Example: You ask "What is the biggest risk?" → "Tenant concentration: one tenant pays 41% of rent." Sources: Rent Roll.

⑤ Credit Memorandum

AI writes the full final report: recommendation, strengths, weaknesses, stress tests and best-fit lenders.

Example: "Approve with Conditions". Stress test: "If profit falls 10%, DSCR becomes 1.31x — still passes." Suggested lender: "Regional Bank – 88% fit".

⑥ Extraction Audit

Shows how well the AI reads documents across all deals, and where humans corrected it.

Example: "312 fields extracted, 87% auto-accepted. 10:42 – analyst changed property value from \$24.0M to \$24.5M."

⑦ Syndication & Term Sheets

Shows loan offers sent to lenders and how big loans are shared between several banks.

Example: "DL-1042: SOFR + 285, 5 yr, status Negotiating." A \$40M loan split: Lead Bank 50%, Bank B 30%, Bank C 20%.

⑧ Covenant Compliance

Watches the promises borrowers made and alerts you when one is close to breaking.

Example: "Minimum DSCR 1.20x — actual 1.08x — **Breach**. Alert: High severity, next test 2026-12-31."

5. What you get

- Loan files reviewed in minutes instead of days.
- Numbers pulled from documents automatically — even from scans and photos.
- Mismatches between documents caught before they become problems.
- A ready-to-present credit report with a clear recommendation.
- Everything saved in a database, so nothing is lost when you close the browser.

6. Good to know

- **No fake data:** if the AI fails, you see a real error and a Retry button — never made-up results.
- **Safe key:** the AI key lives only in the backend, never in the website code.
- **Limits:** files up to 20 MB (scans up to 14 MB). Free AI keys allow only a few requests per minute, so pages may load slowly.
- **Demo deals:** the starting loan list is generated by AI for demonstration; upload real documents to replace numbers.

7. Quick fixes

Problem	Fix
Top bar shows "AI Engine Offline"	Check internet and the key in <code>backend/.env</code> , then restart.
Page says "Deal not found"	The deal list was regenerated — pick a deal again from the Inbox.
Error 429 / slow loading	AI rate limit reached — wait a minute and click Retry.
Scan not readable	Upload a clearer, higher-resolution copy.